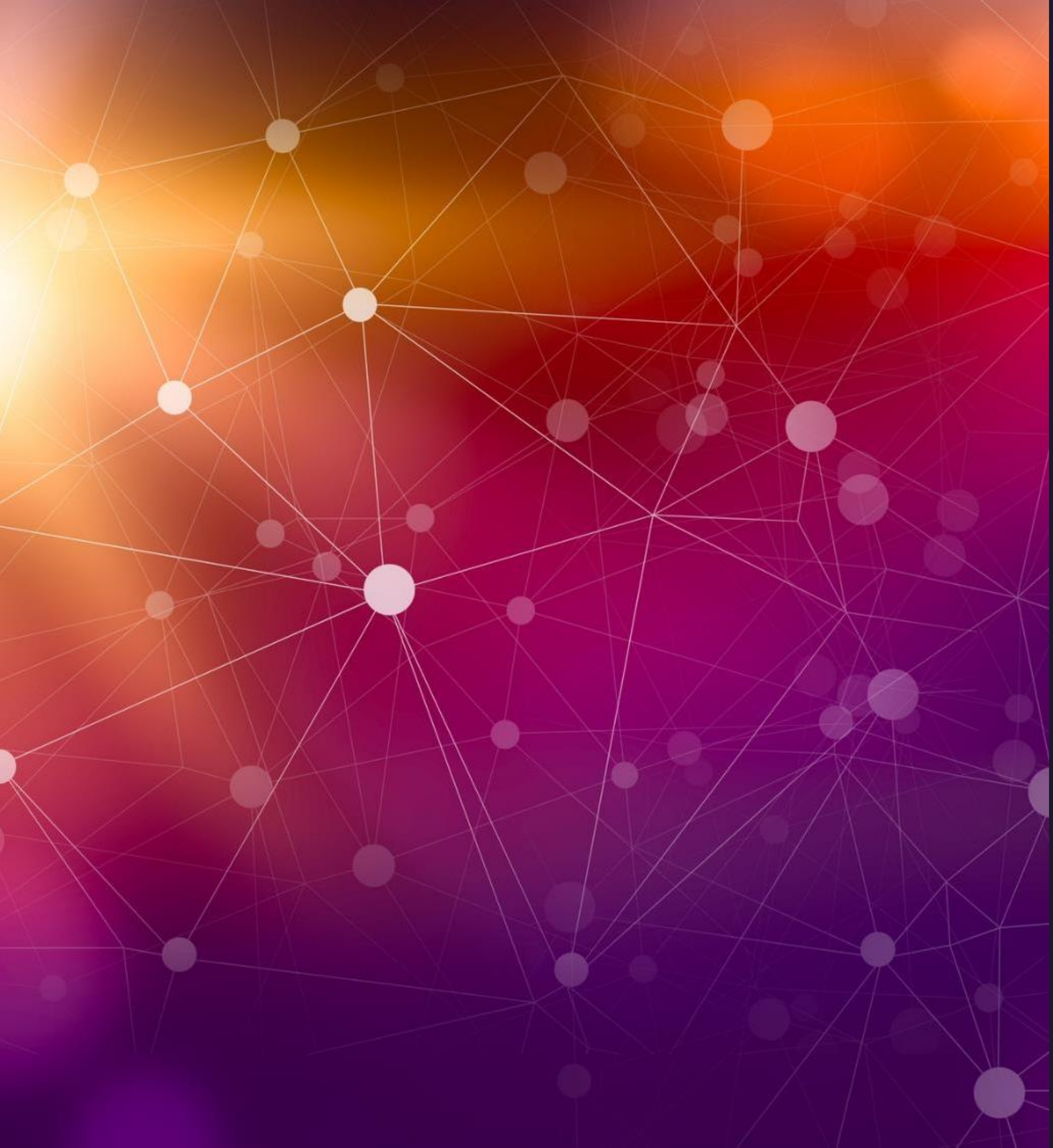


Collision Avoidance Phasing

by Dang H Ngo

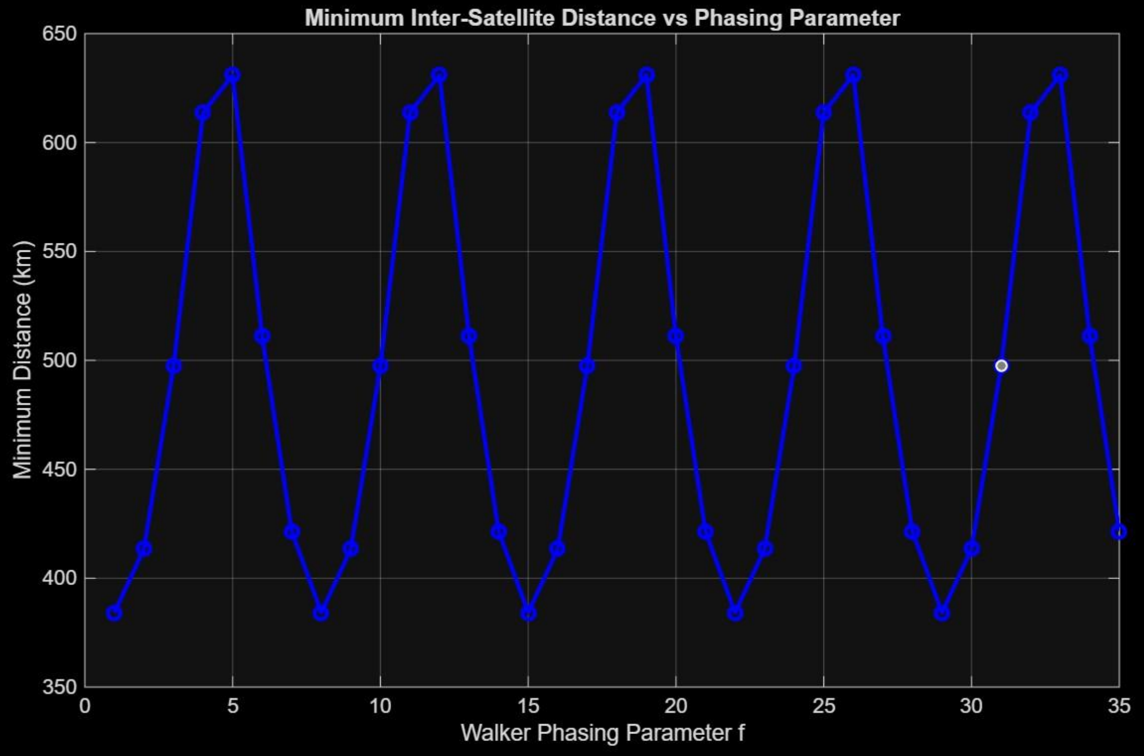
Walker Delta Phasing Protocol

$$M_o^{(p,s^*)} = \frac{360^o}{S} s^* + \frac{360^o(p)(f)}{n} \text{ - (mod) } 360°$$

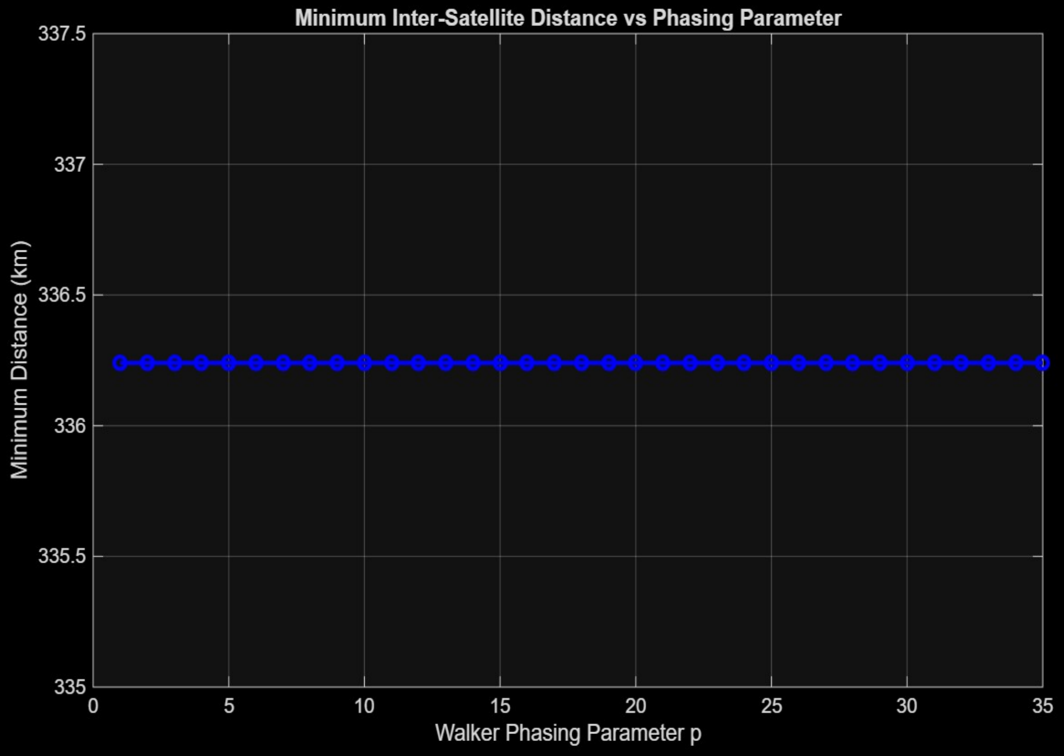
- s^* - Satellite index = {0, 1, 2, …, S – 1}
- p – plane index = {0, 1, 2, …, P – 1}
- n – number of planes
- $\frac{360^o}{S} s^* = \boldsymbol{intrap}\text{lane spacing of satellites}$
- *f* - *Walker Delta phasing integer, controlling interplane time offset*

The Magic '*f*' *parameter*

- RAAN (Ω) – Uniformly spaced across 360°
- omega – Argument of Periapsis fixed at 0 for simplicity

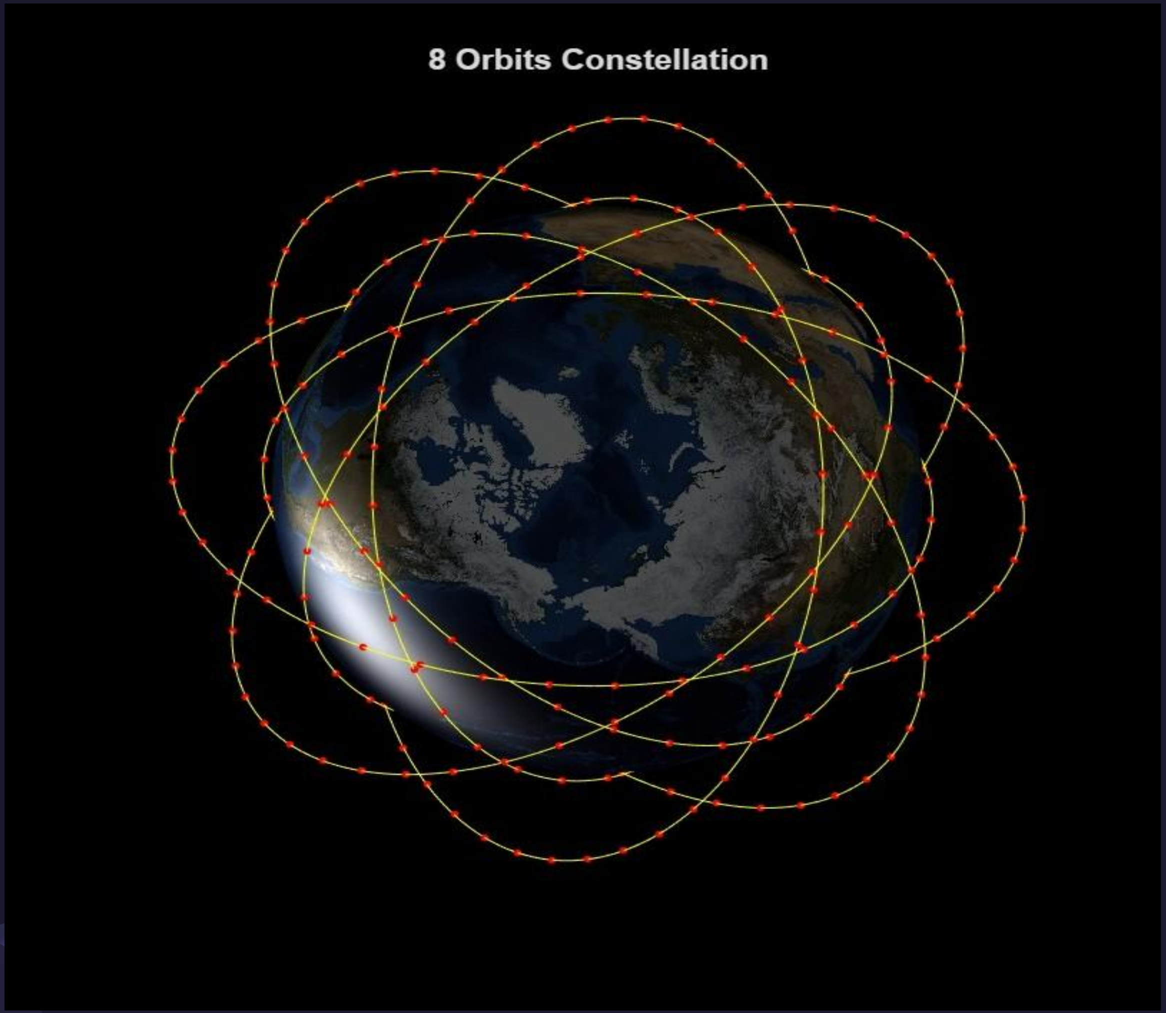


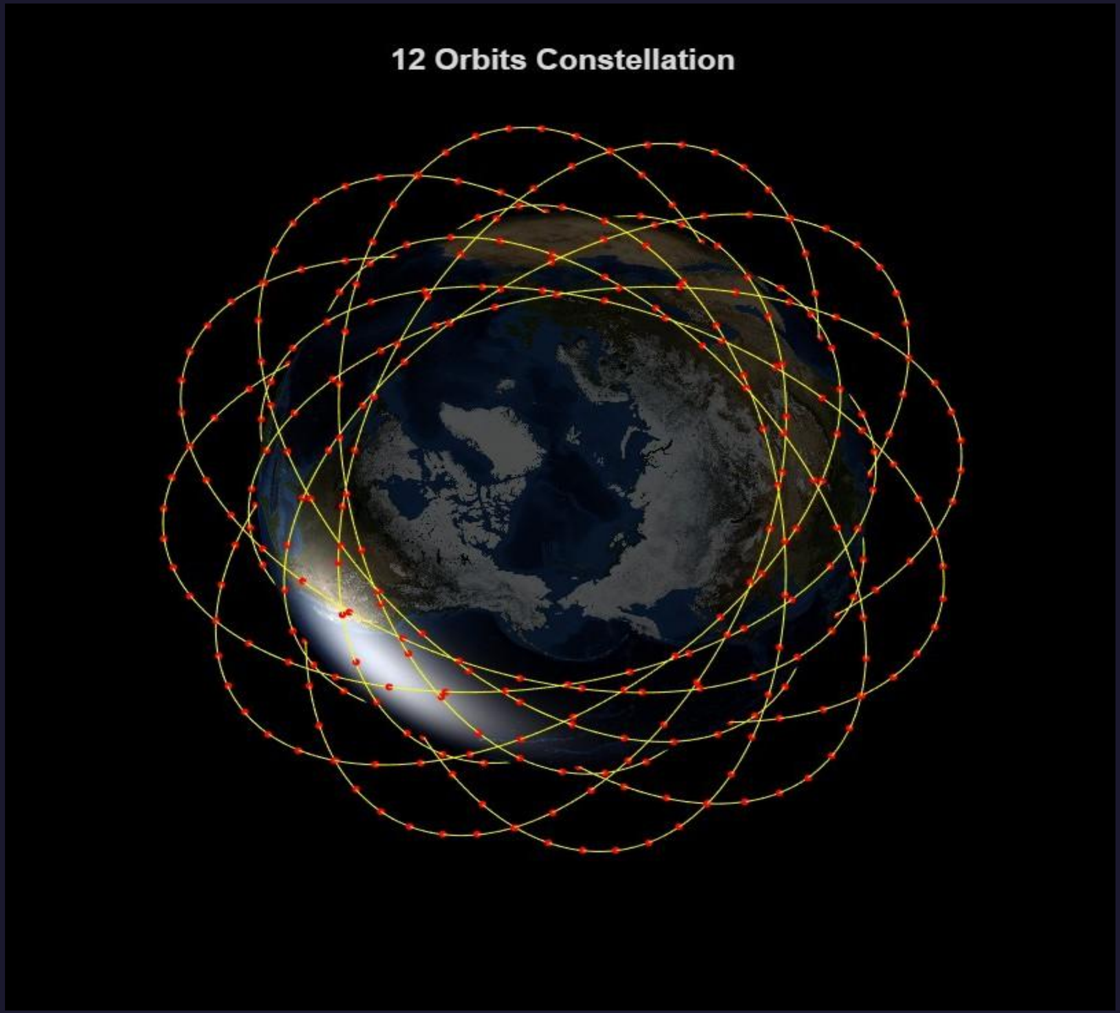
8-Orbit optimal f parameter



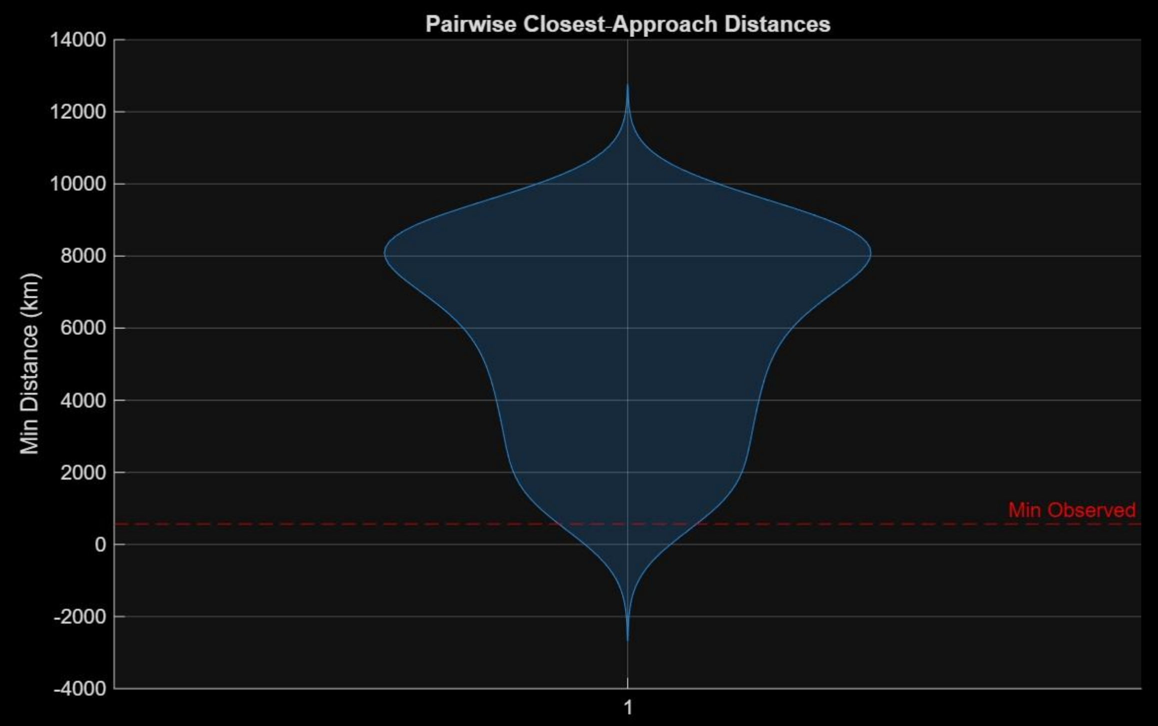
12-Orbit optimal *f* parameter

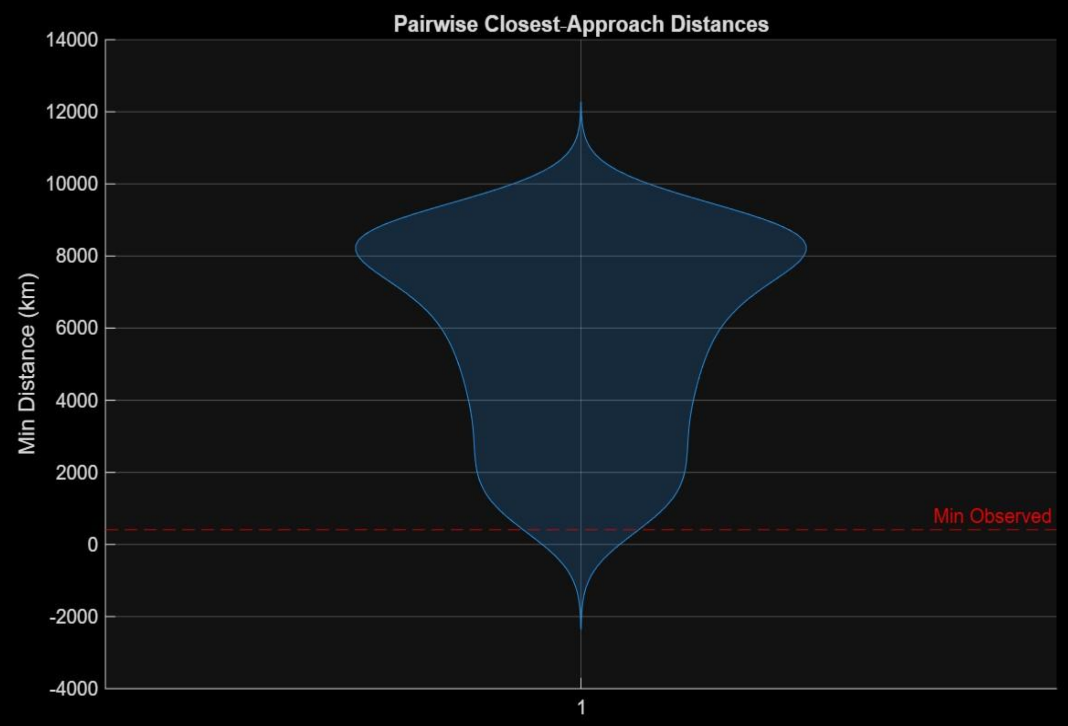
8 orbits and 12 Orbits Constellations





Violin Graphs

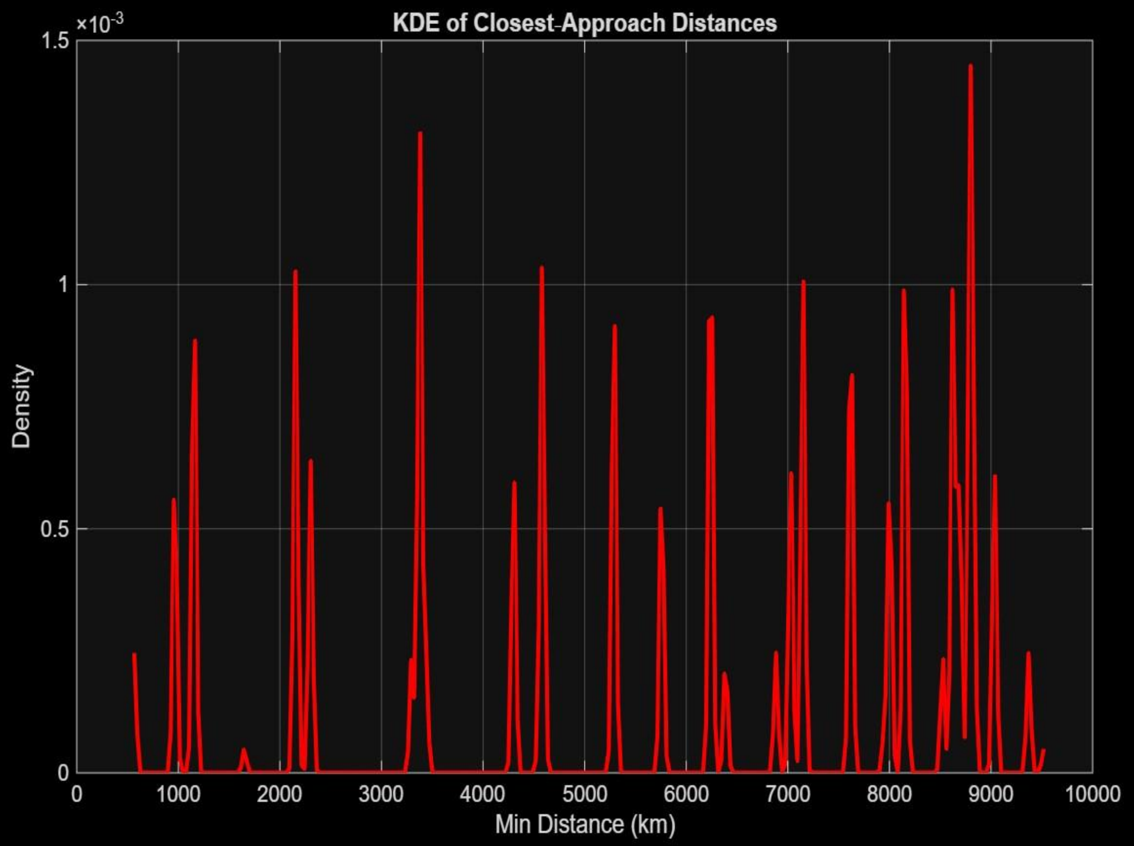


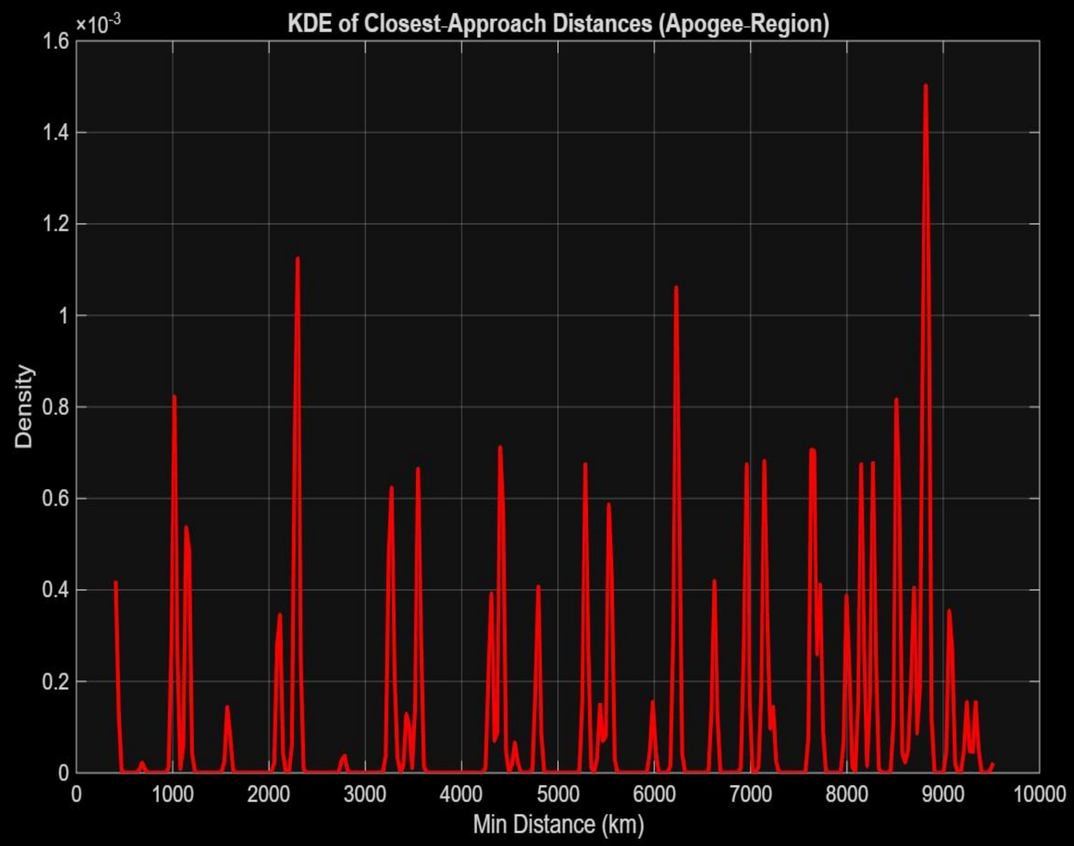


8 orbits

12 orbits

Kernel Density Estimate Graphs





8 orbits

12 orbits

A clockwork analogy

Goal: Spread 36 Satellites in each of P planes so they never collide at crossings

Problem: A collision can only occur if *two* satellites arrive at *the same node point* at *the same instant.*

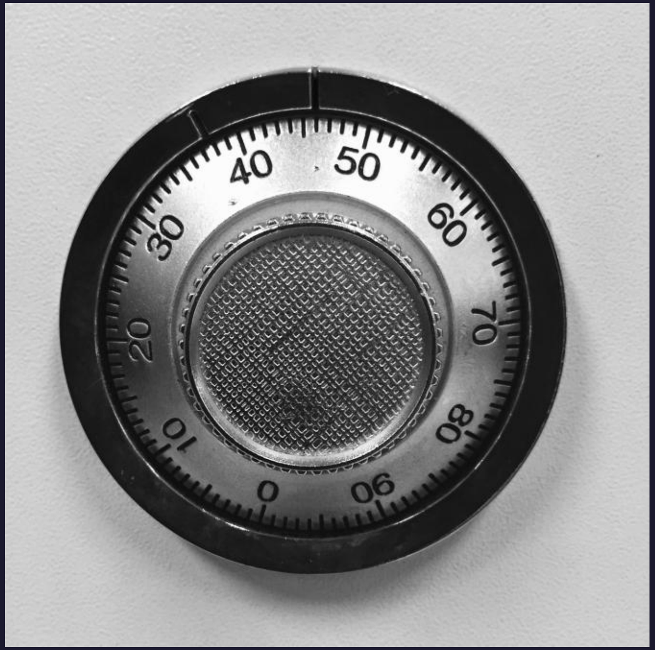
Solution: Each orbital plane has its own ‘clock hand. By giving each plane’s clock a different starting offset, we ensure that adjacent planes’ hands never point to the node at the same instant.

The Key Idea

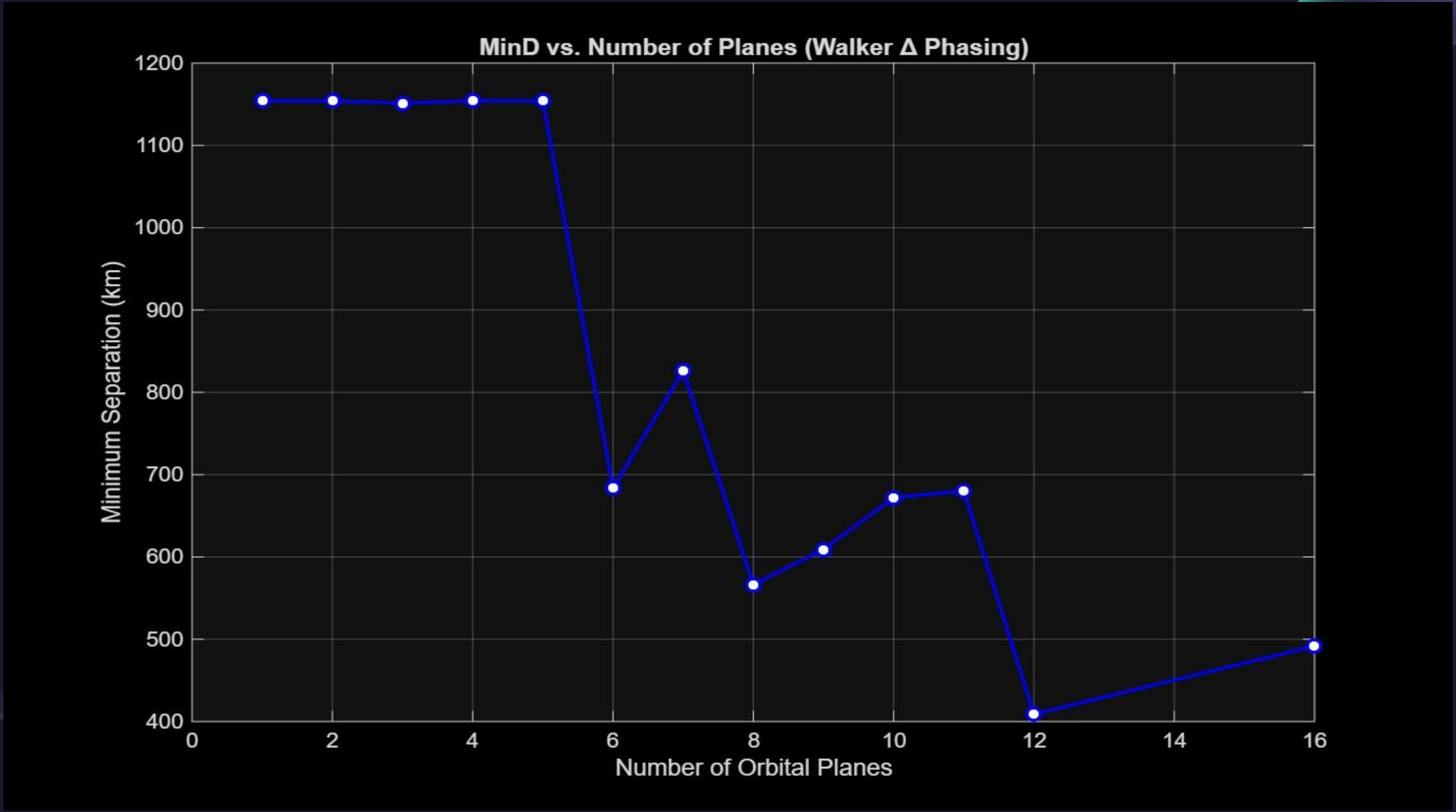
- Imagine 30 clocks (30 planes) arranged in a circle – Each with a single hand
- All hands rotate at the same speed, but we set each one to start at a different mark
- Even though the hands turn together, they never overlap
- The initial start is the Walker phase offset or the ***magic f*** *parameter*
- We offset the starting angle (time), each satellite's path staggered
  - They orbit at the same pace
  - BUT… never pass through the same point at the same time

A Padlock Orange Analogy

- **Rotating dial has numbers 0….60**
- **The pointer shows what number we're pointing to**
- **Inside, there are multiple internal rings (Orbits)**
- **Each ring has its own notch (node crossing)**
- **Notches line up, the lock opens (collision)**
- **The starting angle of ring is the initial true anomaly**
- **Dials rotate at the same speed (Satellites orbiting)**
- **Each plane has a different starting position**
- **Their notches will never align at the same time**



Results of Walker Delta Phasing Simulations



Case Study 3: Elliptical Constellation

- 30 Elliptical Orbital Planes
- 360 Satellites, even spaced in Mean Anomaly
- RAAN Ω varies between adjacent planes by 12 degrees
- Assume constants for all other Keplerian elements
- Perifocal Frame is used to compute Orbital Position
- Transforms to Earth Centered Inertial Frame to compare satellites across planes
- Compute:
  - Satellite position in space over time
  - Minimum safe distance over time

The Key Characteristics of the 10,800 Constellation

- 30 Orbital Planes X 360 Satellites per plane
- Perigee: 200 km Apogee: 2000 km in LEO
- Semi-major axis = 7,271 km
- Eccentricity = .1205
- Angle of inclination *i* = 53^{o}
- *RAAN staggers = 12^{o}*
- *Phasing Parameter: f = 4*
- *Satellites are staggered in true anomaly*

Subset Filtered for Analysis

- **Top 10% Near Apogee Shows the Highest Satellite Density**
  (Satellites cluster where they move slowest)
- **Apogee Is Where Velocity Slows — Clustering Increases Risk**
- **Apogee Becomes the Most Statistically Likely Region for Close Encounters**
- **Walker Δ Phasing Prevents Collisions at Nodes**
  (No two satellites arrive at the node at the same time)
- **Nodes Are Temporally Deconflicted by Design**
- **Apogee Lacks That Temporal Protection**
  (No guarantee that satellites arrive at different times)

The Inner Workings

- **Identify the High-Risk Zone**
  Top 10% of satellites closest to **apogee**, where satellite velocity is slowest and density peaks.

- **Compare Initial Mean Anomalies to Apogee Angle (π)**
  We measured how close each satellite's starting position is to apogee.
  Satellites with $M_0 \approx \pi$ are most likely to linger near apogee.

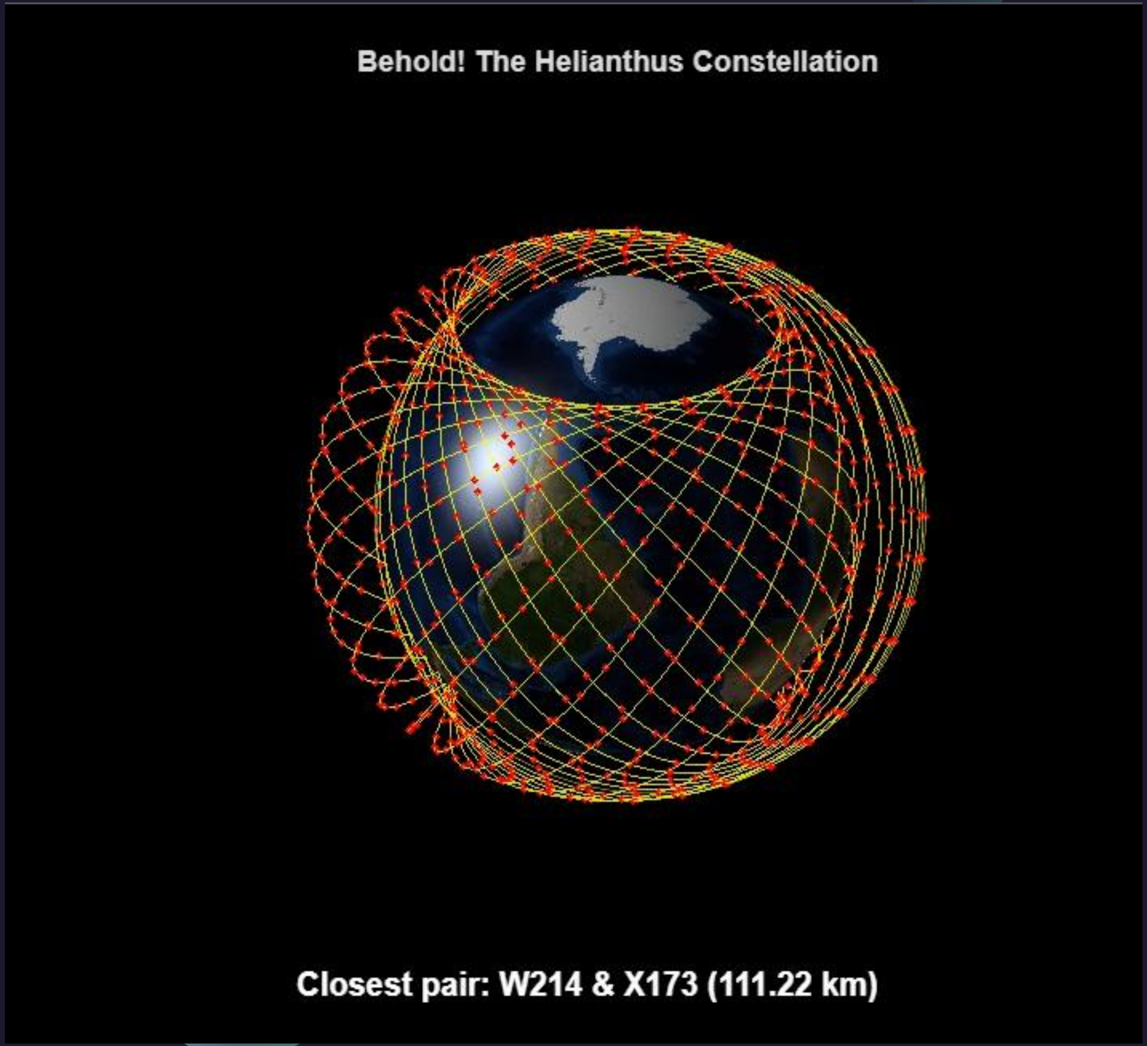
- **Select & Track the Apogee Subset**
  Extracted **1,080 satellites** with mean anomalies closest to π.
  Propagated their **full orbital trajectories**

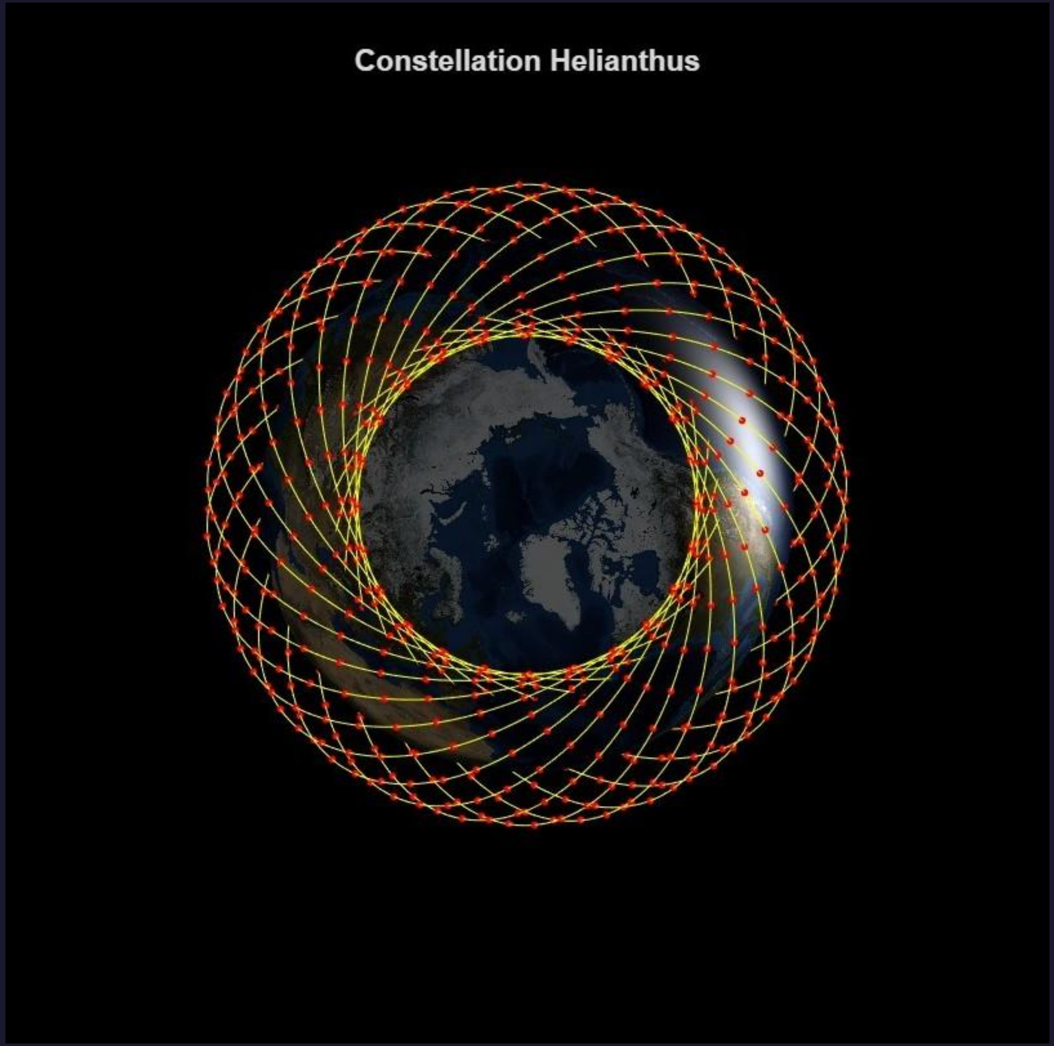
- **Compute Closest Approaches**
  Calculated **all pairwise distances** among the 1,080 satellites over time.
  *Tracked the **minimum Euclidean distance** for each satellite pair.*

- ***Identify the Critical Case***
  *Out of **581,000 total satellite pairs**, we isolated the **single closest encounter**.*

Outcome: Even in the most collision-prone region, the constellation maintained safe separation — validating the effectiveness our phasing algo

Constellation Helianthus





Conclusions and Findings

- **Walker Delta Phasing Effectively Deconflicts Massive Constellations**
- **Node Crossings Were Fully Time-Separated by Design**
  No pair of satellites reached a shared node at the same time — phasing success.
- **Apogee Region Identified as the Most Collision-Prone Zone**
  High satellite density and slow relative motion
- **Simulation Demonstrates Feasibility of Collision-Free, Large-Scale Deployment**
  Results support the safe expansion of next-gen satellite networks using structured phasing

Validated by Simulation of 10,800 Satellites

Helianthus was a rousing success – safely deployed and beautifully phased

Thank you

By

Dang H Ngo

And Jason Lou

References

[1] Math 503A Lecture Notes – Orbital Elements, Satellite Position, Node Crossing

[2] J. R. Wertz, D. F. Everett, and J. J. Puschell, *Space Mission Engineering: The New SMAD*. El Segundo, CA: Microcosm Press, 2011.

[3] J. R. Wertz and W. J. Larson, *Space Mission Analysis and Design*, 3rd ed. Torrance, CA, USA: Microcosm Press, 1999.

[4] Orbital Mechanics – IEEE Spectrum – G Roesler, P Jaffe, and G Henshaw 2017

[5] “Network Performance of LEO Topologies in a High-Inclination Walker Delta Satellite Constellation” – T Royster, J Sun, A Narula-Tam, and T Shake 2023 IEEE Aerospace Conference